

Experience

2023-12 - present	Bioinformatic Scientist <i>Proteovista LLC.</i> - Developed unix-based python scripts to drive product design for genome building technologies and aptamer discovery - Automated data visualization pipelines with Python and Excel VBA for protein-binding DNA arrays, RNA-seq, and NGS - Trained employees on my new tools
2025-06 - 2025-08	Post Graduate Intern <i>Institute of Biomedical and Neural Engineering, Reykjavík, Iceland</i> - Semi-automated ETL-style pipelines for CT segmentation and feature extraction using Mimics API (15x efficiency gain). - Applied missing data imputation (KNNimputer), z-score normalization, feature engineering, and cross-validation to create reproducible supervised ML models for surgical decision support in THA
2024-09 - 2025-03	Graduate Student Research Assistant <i>Biostatistics, Qiongshi Lu Laboratory</i> - Leveraged unsupervised learning of epigenetic data to improve predictive power of biological age estimations - Webscrapped GEO data to control for covariates - Ran ssh remote access PUMAS PRS on Linux Condor's high throughput computing system
2022-06 - 2023-02	Research Specialist <i>Fujifilm Cellular Dynamics Inc</i> - Applied iPSC culture and laboratory techniques to develop robust cell culture processes and assays for iPSC derivation, genetic engineering, and characterization - Wrote protocols, technical reports, SOPs, and documentation
2018-12 - 2020-03	Research Assistant <i>Wisconsin Institute for Discovery</i> - Investigated regional patterning of pluripotent stem cells and characterized various central nervous system tissues - Conducted data collection and authored a grant proposal.
2015-06 - 2021-10	Sailing Instructor <i>Hoofer and Ýmir Sailing Clubs</i> - Directed large-scale recruiting events, coordinating 3–6 instructors and managing up to 40 students/lesson, introducing hundreds to windsurfing. - One of 18 to have ever earned the club's highest designation of trust (Lake Leader).

Education

2023-09 - 2025-08	Biomedical Data Science, M.S. <i>University of Wisconsin - Madison</i>
2017-09 - 2022-05	Biomedical Engineering, Neurobiology, B.S. <i>University of Wisconsin, Madison</i>

Noah Nicol



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Programming

Python (NumPy, Pandas, Biopython, SciPy)
R (RStudio, Tidyverse), MySQL, MS Visual Basic
Linux/Unix environments
Reproducible Workflows (R Markdown, Jupyter Notebooks, version control/GitHub)

Bioinformatics & DS:

Machine Learning (Scikit-learn, TensorFlow, Neural Networks)

Data Visualization (Tableau, Matplotlib, Seaborn, ggplot2, Excel)

High-throughput Sequencing Data Analysis (NGS, RNA-seq)

Statistical Modeling (ANOVA, Regression, PCA, Clustering)

Data Wrangling and Prompt Engineering

Publications

Modular derivation of diverse, regionally discrete human posterior CNS neurons enables discovery of transcriptomic patterns

Science Advances, 2022

<https://doi.org/10.1126/sciadv.abn7430>

7615 Genomic Consequences of GRHL2 Overexpression in ER+ Breast Cancer Cells

Journal of the Endocrine Society, 2024

<https://doi.org/10.1210/jendso/bvae163.1809>

Personal Projects

HuckHub

Location-based PWA for matching frisbee throwing partners. Built with Next.js 15, Supabase (PostgreSQL/PostGIS), real-time push notifications, and geospatial matching algorithms.

Ymir Safety System

Built a PWA (Astro, React, Tailwind, Postgres) to automate boat checkouts, safety alerts, maintenance tracking, and bilingual member management using a QR-powered workflow, UTC-safe scheduling system with real-time overdue detection,

Personal AWS IoT Sandbox Project

Experimented with AWS IoT Core and Python (boto3) to prototype MQTT message publishing, IoT device shadow updates, and secure certificate-based communication between simulated edge devices and the AWS cloud.

Volunteering

2021-08 - present	<i>Hoofer Sailing Club Outreach Coordinator</i> I share my love of windsurfing, ice kiting, and sailing with club members and UW students
2018-04 - present	<i>Ultimate Frisbee</i> • I organize youth ultimate day camps • Assistant coach for Madison West High School's varsity team • Founded two adult Madison annual club teams • Teach at Madison Radicals sponsored youth camps

Laboratory Techniques

iPSC Cell Culture, ICC, IHC, Flow Cytometry, Confocal Microscopy, RT-qPCR

Certificates

2023-02

Brainest Python Developer Training Program

2025-03

Data or Specimens Only Research